

Operating Systems (CS2018)

Sessional-II Exam

Date: April 9th 2025

Course Instructor(s)

Dr. Afzal Ahmed

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 3

Roll No

Section

Student Signature

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Attempt all the questions.

CLO 3: Analyze memory management techniques.

Q1: Find out the number of page fault, hit ratio, and miss ratio if there are three page frames, using the following page replacement algorithms:

[10 Marks]

- 1) FIFO
- 2) Optimal
- 3) LRU

for the following page reference string, $\underline{5}, \underline{0}, 1, 0, \underline{2}, \underline{3}, 0, 2, 4, 3, \underline{3}, \underline{2}, 0, 2, 1, 2, 7, 0, 1, 1, 0$.

CLO #: 3 Analyze memory management techniques.

Q2: Consider a swapping system in which memory consists of the following hole sizes in memory order: 10 MB, 4 MB, 20 MB, 18 MB, 7 MB, 9 MB, 12 MB, and 15 MB.

[10 Marks]

Investigate which hole is taken for successive segment requests of

- (a) 12 MB
- (b) 10 MB
- (c) 9 MB

for first fit? Now repeat the question for best fit, worst fit, and next fit.

CLO #: 3 Analyze memory management techniques.

Q3a: Differentiate between multilevel page table and inverted page table.

[5 Marks]

Q3b: Explain shared pages and shared libraries as the design issues for paging systems.

[5 Marks]